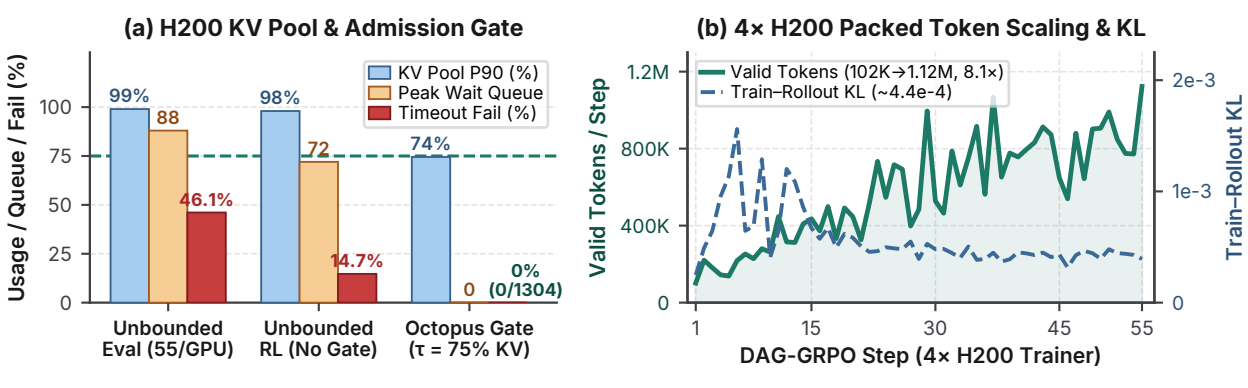
(a) H200 KV Pool & Admission Gate
KV Pool P90 (%)
Peak Wait Queue
Timeout Fail (%)
Usage / Queue / Fail (%)
100
75
50
25
0
99%
88
46.1%
98%
72
14.7%
74%
0
0%
(0/1304)
Unbounded
Eval (55/GPU)
Unbounded
RL (No Gate)
Octopus Gate
(τ = 75% KV)
(b) 4× H200 Packed Token Scaling & KL
Valid Tokens (102K→1.12M, 8.1×)
Train–Rollout KL (~4.4e-4)
Valid Tokens / Step
1.2M
800K
400K
0
Train–Rollout KL
2e-3
1e-3
0
1
15
30
45
55
DAG-GRPO Step (4× H200 Trainer)